

Electromagnetic Induction Problem Generator

Hide Teacher Mode

Faraday's Law Analysis: $\vec{B}(t) \rightarrow \Phi(t) \rightarrow e(t) \rightarrow i(t)$

Coil Parameters

Turns (N)

10

Radius r (cm)

10 cm

Resistance R (Ω)

0.5 Ω

Problem Statement

A circular coil with **10** turns and radius **10 cm** is placed in a uniform magnetic field $\vec{B}$. The total resistance of the circuit is **0.5 Ω** .

The vector $\vec{B}$ is perpendicular to the plane of the coil. Its magnitude varies with time as shown in the graph.

- $t = 0 \rightarrow 1\text{s}$: Linear increase (Rising Phase)
- $t = 1 \rightarrow 3\text{s}$: Constant Field (Plateau)
- $t = 3 \rightarrow 4\text{s}$: Linear decrease (Falling Phase)

Questions:

- Determine the direction of induced current in each phase (Lenz's Law).
- Calculate the Flux Φ at $t = 0.5\text{s}$ and $t = 3.5\text{s}$.
- Calculate the induced EMF (e) and current (i) for the rising and falling phases.

Solution Key

1. Phase I (Rising)

Slope dB/dt : 0.80 T/s

EMF $e = -NS(dB/dt)$: -0.251 V

Current $i = e/R$: -0.503 A

2. Phase II (Constant)

Slope dB/dt : 0 T/s

EMF: 0 V

3. Phase III (Falling)

Slope dB/dt : -1.20 T/s

EMF e : 0.377 V

Current i : 0.754 A

